

External Interrupt

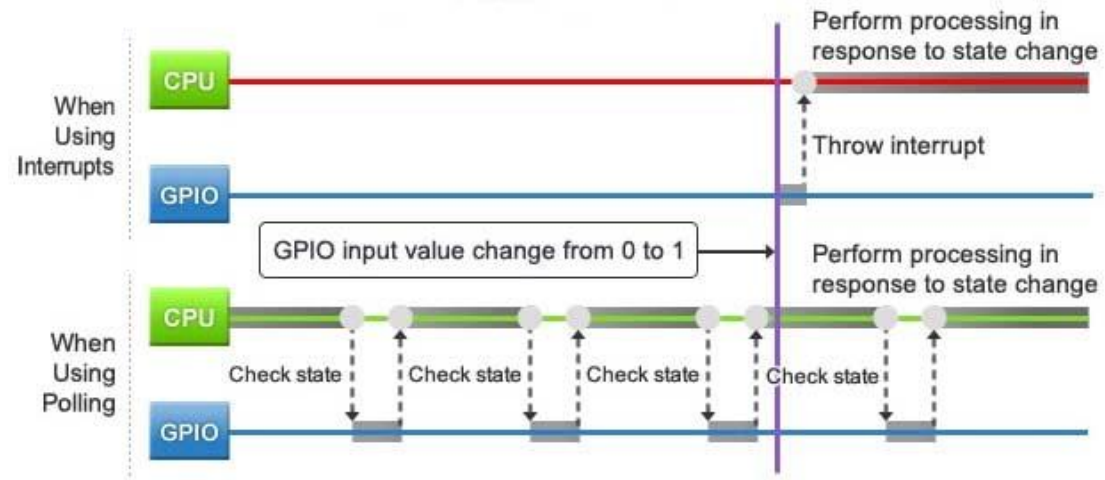
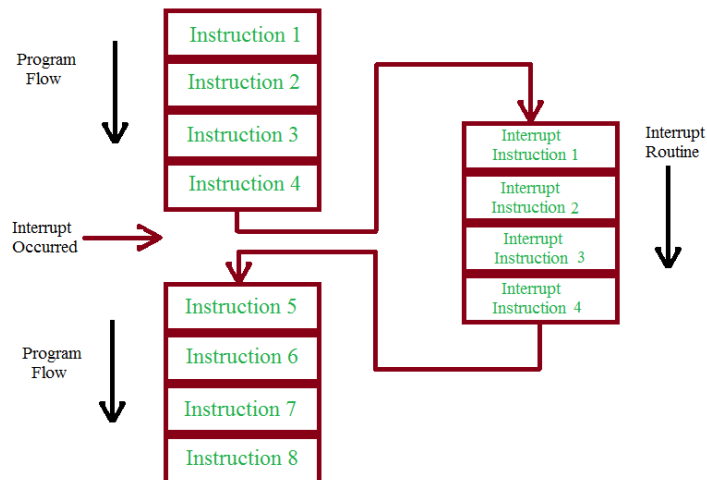


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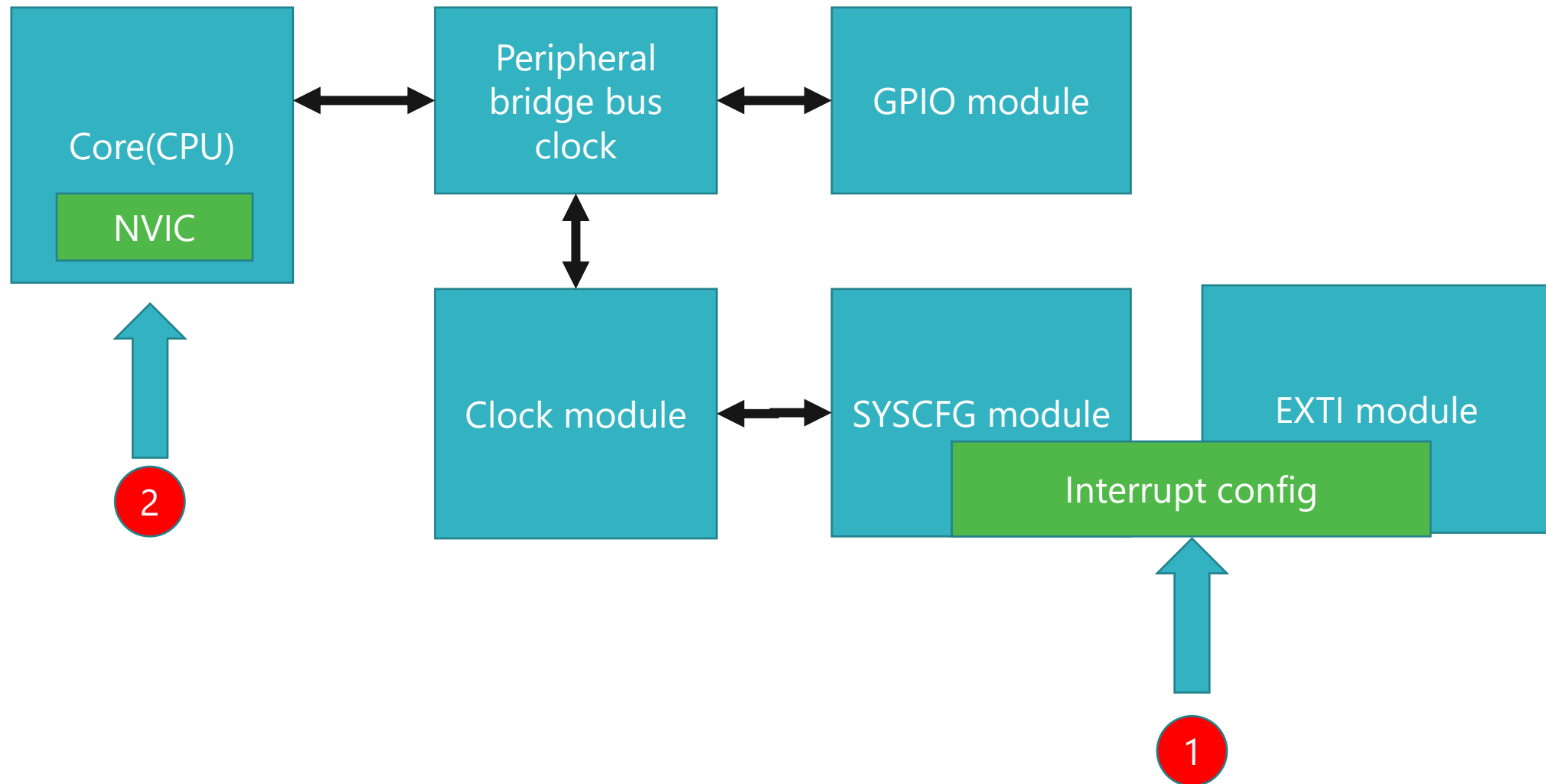
- 1. Polling and interrupt**
- 2. System configuration controller (SYSCFG)**
- 3. External interrupt/event controller (EXTI)**
- 4. Configuration**

1. Polling and interrupt

Criteria	Polling	Interrupt
Concept	Continuously checking device status	Notifying when an event occurs
Operation	The microcontroller runs a loop to check status	The microcontroller pauses the current program and executes the ISR
Advantages	- Simple and easy to implement - Easy to understand and control	- More efficient CPU usage - Frees up CPU resources for other tasks
Disadvantages	- Time-consuming and resource - Can reduce system performance	- More complex management - Potential for interrupt conflicts
Use Cases	When requirements are simple or real-time is not critical	When fast response and high performance are needed

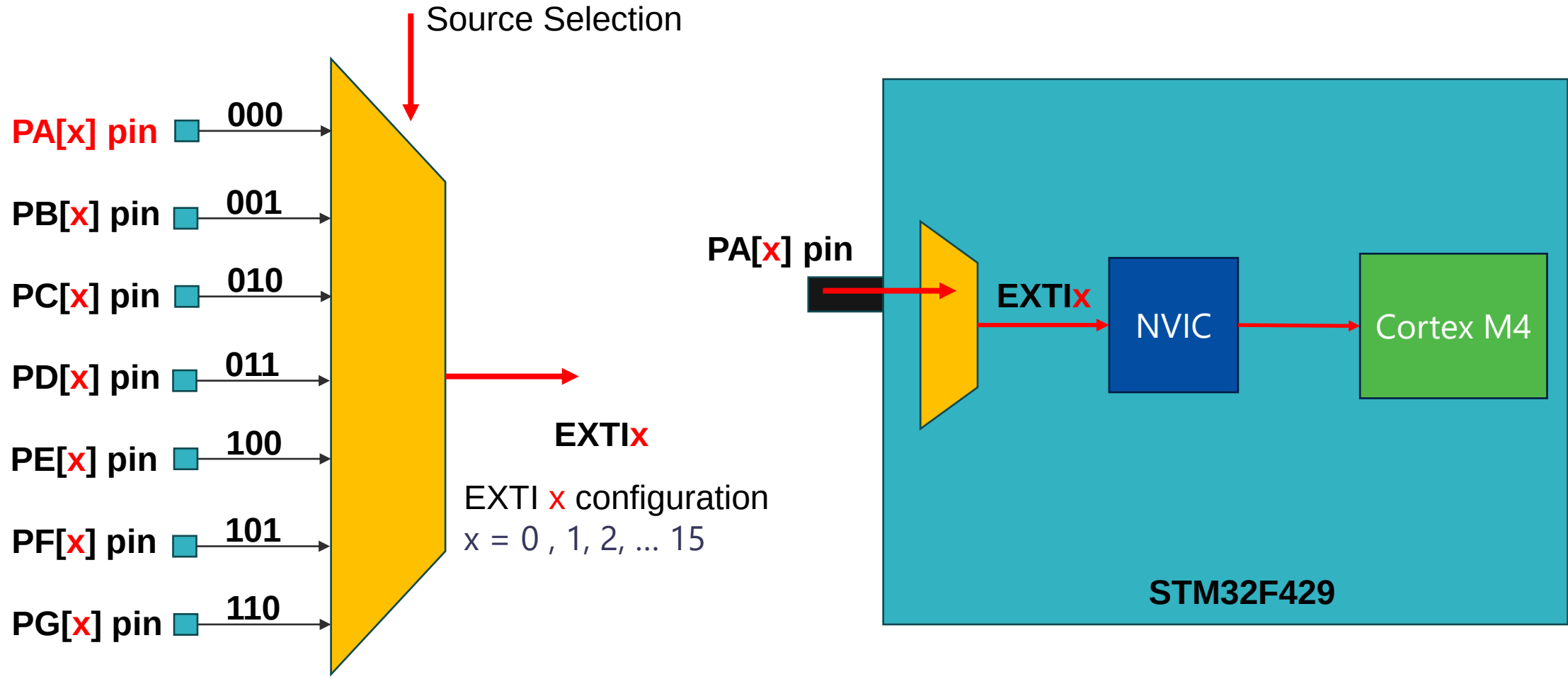


1.1. Modules overview



2. System configuration controller (SYSCFG)

SYSCFG external interrupt configuration register (SYSCFG_EXTICR1)



One mux for each of EXTI0, EXTI1, .. EXTI15

2. System configuration controller (SYSCFG)

RCC APB2 peripheral clock enable register (RCC_APB2ENR)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved					LTDC EN	Reserved			SAI1EN	SPI6EN	SPI5EN	Res.	TIM11 EN	TIM10 EN	TIM9 EN
					rw				rw	rw	rw		rw	rw	
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reser- ved	SYSCF G EN	SPI4E N	SPI1 EN	SDIO EN	ADC3 EN	ADC2 EN	ADC1 EN	Reserved	USART 6 EN	USART 1 EN	Reserved	TIM8 EN	TIM1 EN		
	rw	rw	rw	rw	rw	rw	rw		rw	rw		rw	rw		

Bit 14 **SYSCFGEN**: System configuration controller clock enable

This bit is set and cleared by software.

0: System configuration controller clock disabled

1: System configuration controller clock enabled

2. System configuration controller (SYSCFG)

SYSCFG external interrupt configuration register x (SYSCFG_EXTICRx)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
EXTI3[3:0]				EXTI2[3:0]				EXTI1[3:0]				EXTI0[3:0]			
RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW

Bits 15:0 **EXTIx[3:0]**: EXTI x configuration (x = 0 to 3)

These bits are written by software to select the source input for the EXTIx external interrupt.

0000: PA[x] pin

0001: PB[x] pin

0010: PC[x] pin

0011: PD[x] pin

0100: PE[x] pin

0101: PF[x] pin

0110: PG[x] pin

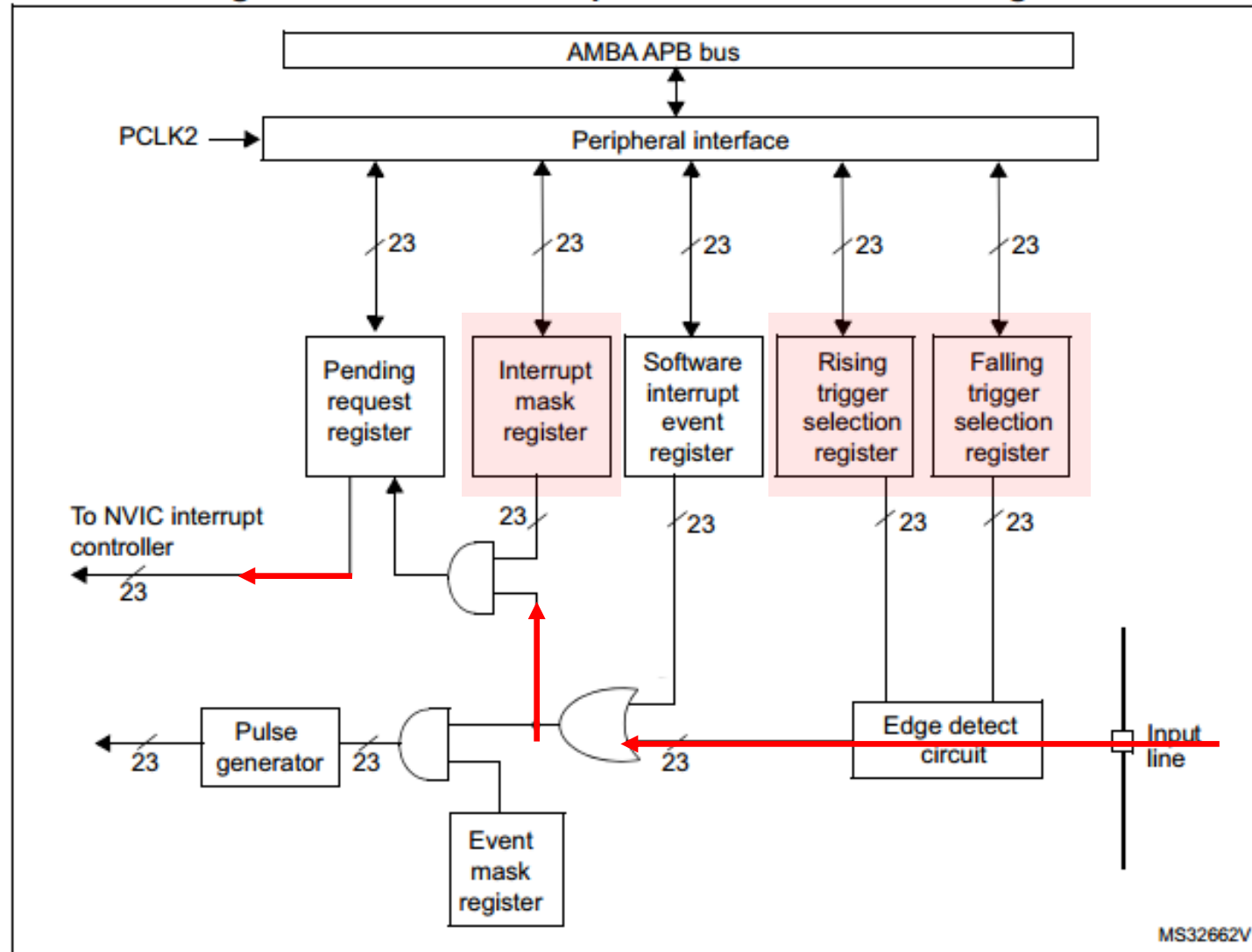
3. External interrupt/event controller (EXTI)

The main features of the EXTI controller are the following:

- Independent **trigger** and **mask** on each interrupt/event line
- Dedicated **status bit** for each interrupt line
- Generation of up to **23** software event/interrupt requests
- Detection of external signals with a pulse width lower than the APB2 clock period.

3. External interrupt/event controller (EXTI)

External interrupt/event controller block diagram:

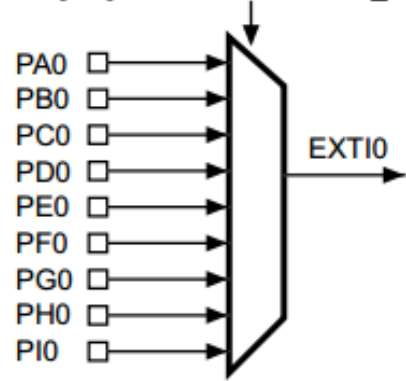


3. External interrupt/event controller (EXTI)

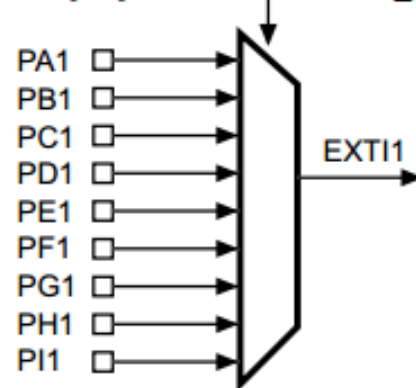
External interrupt/event line mapping:

GPIOs are connected to the 16 external interrupt/event lines in the following manner:

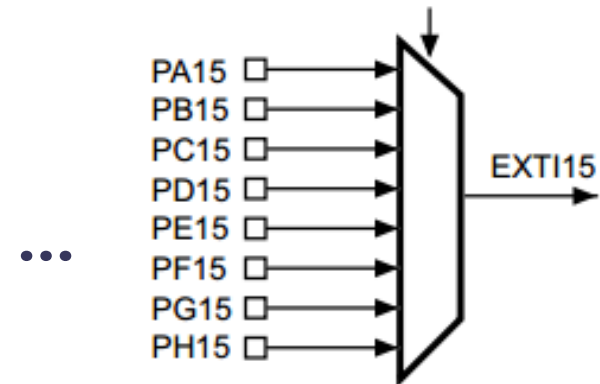
EXTI0[3:0] bits in the SYSCFG_EXTICR1 register



EXTI1[3:0] bits in the SYSCFG_EXTICR1 register



EXTI15[3:0] bits in the SYSCFG_EXTICR4 register

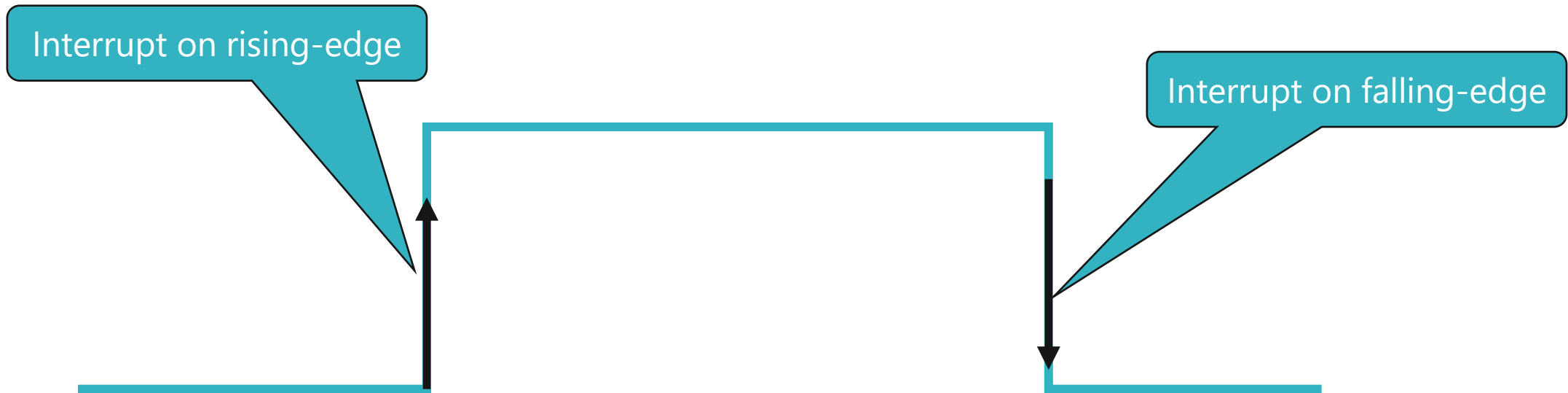


The seven other EXTI lines are connected as follows:

- EXTI line 16 is connected to the PVD output
- EXTI line 17 is connected to the RTC Alarm event
- EXTI line 18 is connected to the USB OTG FS Wakeup event
- EXTI line 19 is connected to the Ethernet Wakeup event
- EXTI line 20 is connected to the USB OTG HS (configured in FS) Wakeup event
- EXTI line 21 is connected to the RTC Tamper and TimeStamp events
- EXTI line 22 is connected to the RTC Wakeup event

3. External interrupt/event controller (EXTI)

Trigger selection:



3. External interrupt/event controller (EXTI)

Rising trigger selection register(EXTI_RTSR)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved									TR22	TR21	TR20	TR19	TR18	TR17	TR16
									r/w	r/w	r/w	r/w	r/w	r/w	r/w
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TR15	TR14	TR13	TR12	TR11	TR10	TR9	TR8	TR7	TR6	TR5	TR4	TR3	TR2	TR1	TR0
r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w

Bits 22:0 **TRx**: Rising trigger event configuration bit of line x

0: Rising trigger disabled (for Event and Interrupt) for input line

1: Rising trigger enabled (for Event and Interrupt) for input line

Falling trigger selection register (EXTI_FTSR)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved									TR22	TR21	TR20	TR19	TR18	TR17	TR16
									r/w	r/w	r/w	r/w	r/w	r/w	r/w
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TR15	TR14	TR13	TR12	TR11	TR10	TR9	TR8	TR7	TR6	TR5	TR4	TR3	TR2	TR1	TR0
r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w

Bits 22:0 **TRx**: Falling trigger event configuration bit of line x

0: Falling trigger disabled (for Event and Interrupt) for input line

1: Falling trigger enabled (for Event and Interrupt) for input line.

3. External interrupt/event controller (EXTI)

Interrupt mask register (EXTI_IMR)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved									MR22	MR21	MR20	MR19	MR18	MR17	MR16
									rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MR15	MR14	MR13	MR12	MR11	MR10	MR9	MR8	MR7	MR6	MR5	MR4	MR3	MR2	MR1	MR0
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 22:0 **MRx**: Interrupt mask on line x

0: Interrupt request from line x is masked

1: Interrupt request from line x is **not masked (enable)**

3. External interrupt/event controller (EXTI)

Pending register (EXTI_PR)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved									PR22	PR21	PR20	PR19	PR18	PR17	PR16
									rc_w1	rc_w1	rc_w1	rc_w1	rc_w1	rc_w1	rc_w1
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PR15	PR14	PR13	PR12	PR11	PR10	PR9	PR8	PR7	PR6	PR5	PR4	PR3	PR2	PR1	PR0
rc_w1	rc_w1	rc_w1	rc_w1	rc_w1	rc_w1	rc_w1	rc_w1	rc_w1	rc_w1	rc_w1	rc_w1	rc_w1	rc_w1	rc_w1	rc_w1

Bits 22:0 **PRx**: Pending bit

0: No trigger request occurred

1: selected trigger request occurred

This bit is set when the selected edge event arrives on the external interrupt line.

This bit is cleared by programming it to '1'.

→ Clear the flag immediately in the ISR after confirming that the event has occurred and has been processed.

3. External interrupt/event controller (EXTI)

Pending register (EXTI_PR)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved									PR22	PR21	PR20	PR19	PR18	PR17	PR16
									rc_w1	rc_w1	rc_w1	rc_w1	rc_w1	rc_w1	rc_w1
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PR15	PR14	PR13	PR12	PR11	PR10	PR9	PR8	PR7	PR6	PR5	PR4	PR3	PR2	PR1	PR0
rc_w1	rc_w1	rc_w1	rc_w1	rc_w1	rc_w1	rc_w1	rc_w1	rc_w1	rc_w1	rc_w1	rc_w1	rc_w1	rc_w1	rc_w1	rc_w1

Bits 22:0 **PRx**: Pending bit

0: No trigger request occurred

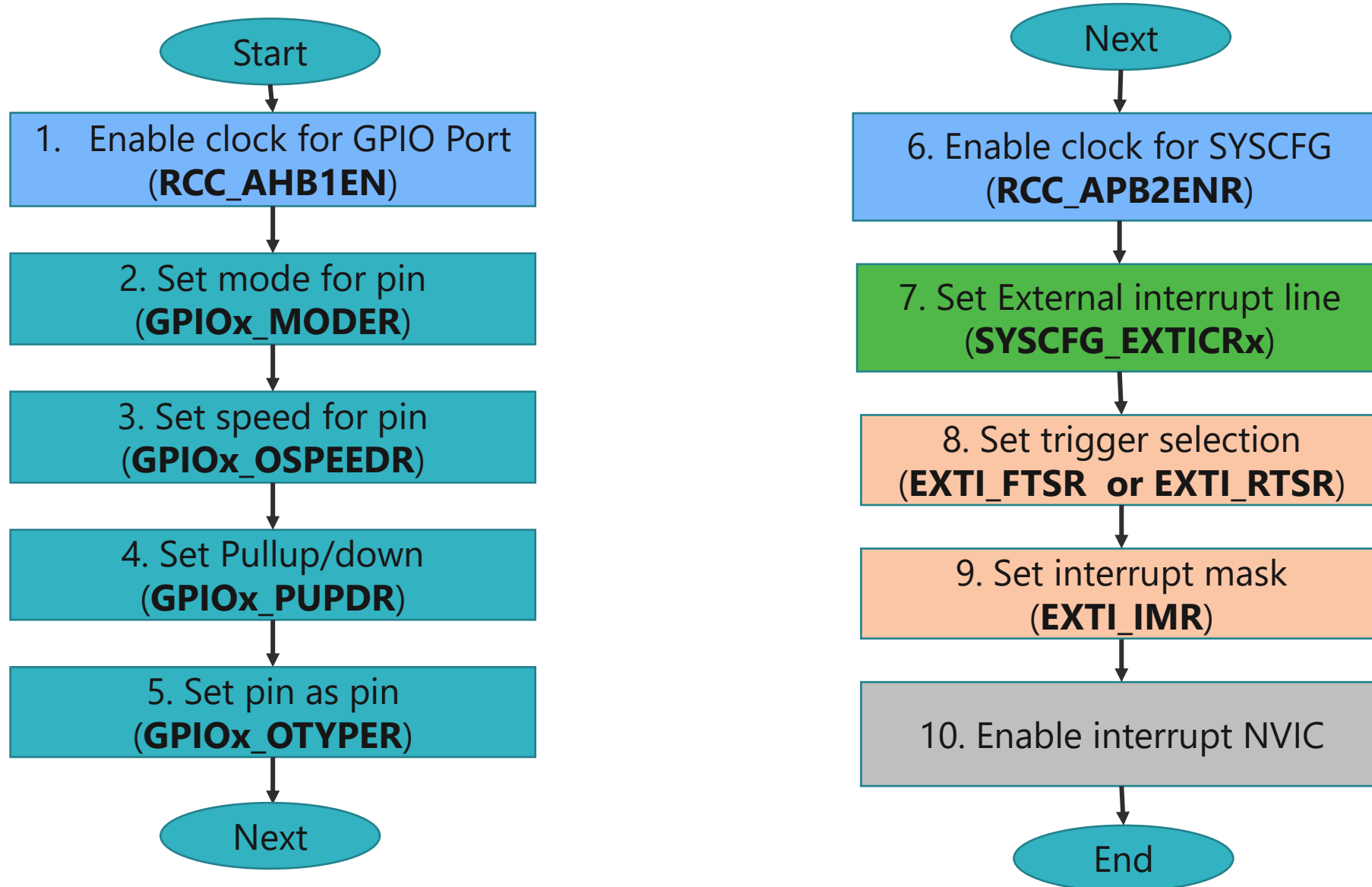
1: selected trigger request occurred

This bit is set when the selected edge event arrives on the external interrupt line.

This bit is cleared by programming it to '1'.

→ Clear the flag immediately in the ISR after confirming that the event has occurred and has been processed.

4. Configuration



4.1. ISR (Interrupt Service Routine)

- ISR (Interrupt Service Routine) is a function or code segment that is executed when an interrupt occurs.
- When an interrupt occurs, the microcontroller temporarily pauses the current program execution and switches to executing the ISR to handle the related event.

The main characteristics of an ISR:

- ❖ Quick execution: The ISR should execute quickly to avoid long interruptions of the main process.
- ❖ Interrupts disabled: Interrupts are typically disabled during ISR execution to prevent conflicts.
- ❖ Interrupt flag: After processing, the interrupt flag is usually cleared to ensure the interrupt is not handled again immediately.

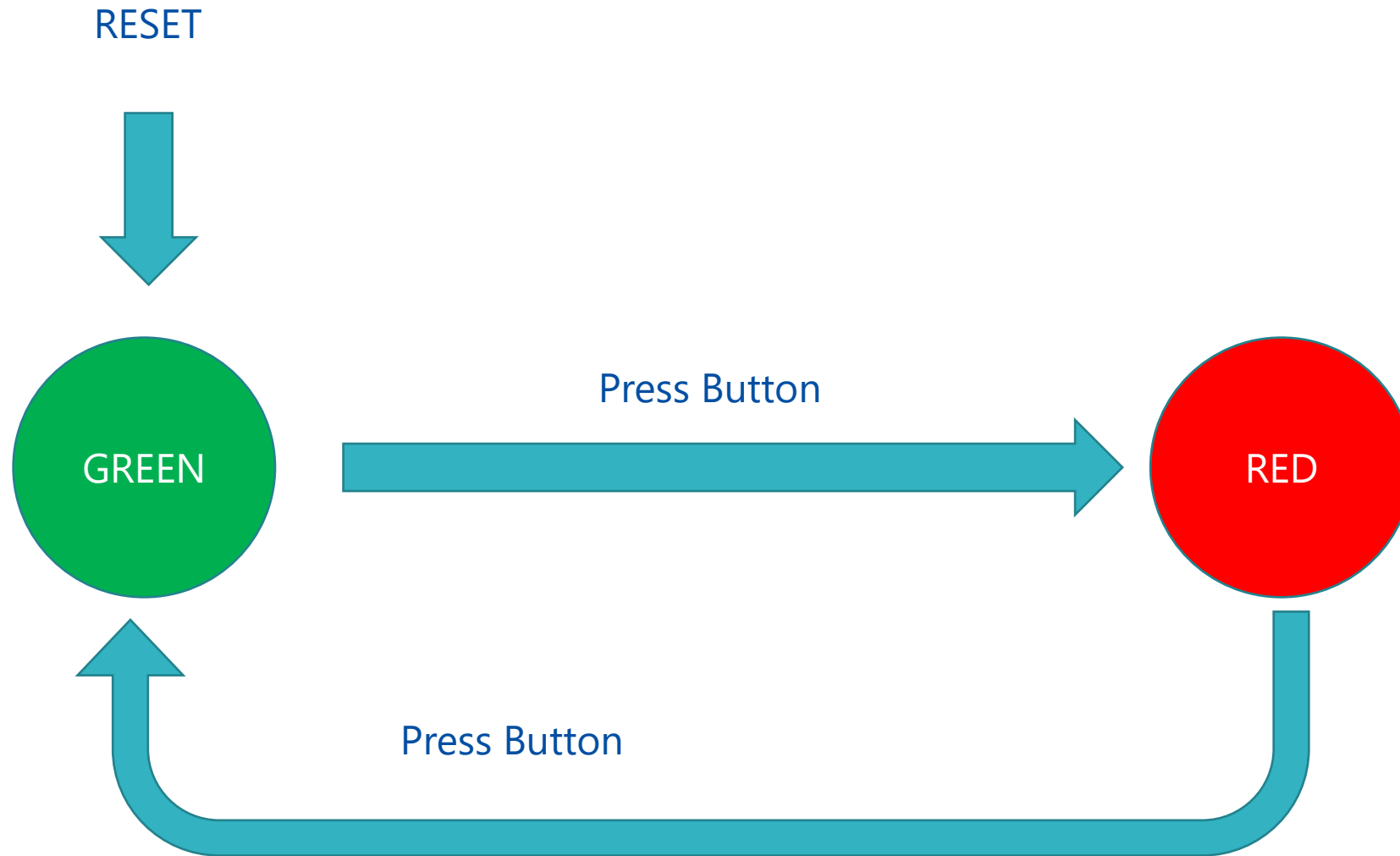
Keyword: Decoupling of interrupt

4.1. ISR (Interrupt Service Routine)

- The ISR (Interrupt Service Routine) for external interrupts is declared in the file `startup_stm32f429xx.s`

No.	EXTIx	Name ISR
1	EXTI0	EXTI0_IRQHandler
2	EXTI1	EXTI1_IRQHandler
3	EXTI2	EXTI2_IRQHandler
4	EXTI3	EXTI3_IRQHandler
5	EXTI4	EXTI4_IRQHandler
6	EXTI5→EXTI9	EXTI9_5_IRQHandler
7	EXTI10→EXTI15	EXTI15_10_IRQHandler

4.2. Assignment 1



4.3. Assignment 2

Setting external interrupt for PB1 pin with the following conditions:

- Input pull-down
- Falling trigger selection

4.4. Assignment 3

Setting external interrupt for PC6 pin with the following conditions:

- Input pull-up
- Falling and rising trigger selection

A nighttime cityscape featuring a prominent skyscraper with a spire, illuminated against a dark sky with a pinkish-orange sunset glow. The city lights reflect on the water in the foreground. A large, semi-transparent, stylized letter 'R' is overlaid on the right side of the image, framing the central skyscraper. The text 'Thank you' is written in white, sans-serif font on the left side of the image.

Thank you